Artificial Intelligence

Revision Sheet

Every topic on its own page • method, worked example, traps

How to use this sheet

- **One topic per page.** Nothing splits across pages — read a page, close the book, and try to reproduce the formula box from memory.
- **Read in this order on each page:** the italic line (what the topic *is*) → the formula box → "how to approach" → the worked example → the traps.
- **The dashed "My notes" box is yours.** Add the variant that caught you out, a formula your coaching notes state differently, or the page number of a question you got wrong.
- **Traps in the orange box are the marks people actually lose.** Re-read only those the night before the exam.

Contents

1. Uninformed Search
2. Informed & Heuristic Search
3. Adversarial Search & Games
4. Propositional Logic
5. First-Order (Predicate) Logic
6. Reasoning under Uncertainty

1. Uninformed Search

Blind strategies with no domain knowledge. Everything is decided by one properties table.

KEY FORMULAS

Costs (b branching, d goal depth, m max depth)

BFS : $O(b^d)$ time and space; DFS : $O(b^m)$ time, $O(bm)$ space

Nodes expanded by BFS

$$1 + b + b^2 + \cdots + b^d$$

WHAT YOU MUST KNOW

- **Uninformed**: BFS, DFS, uniform-cost, iterative deepening, depth-limited. **Informed**: A*, greedy (they use a heuristic).
- BFS is complete and optimal *when step costs are equal*; **UCS** (lowest $g(n)$) is optimal for any non-negative costs.
- DFS is **not complete** on infinite or cyclic spaces, and never optimal — but its memory is only linear.
- **IDDFS** gets DFS's memory with BFS's completeness and optimality, at a small repeated-work cost.

HOW TO APPROACH A QUESTION

1. Identify the frontier structure: queue (BFS), stack (DFS), priority queue (UCS/A*).
2. For "how many nodes expanded", simulate literally, respecting the stated expansion order.
3. Watch whether repeated states are re-added — the question usually says.

WORKED EXAMPLE

Q. Which of BFS, DFS, UCS, A* is *not* uninformed?

A* evaluates $f = g + h$ and h is a heuristic estimate of the remaining cost:

A* Search

WATCH OUT

- Calling depth-limited search "informed" — it just caps the depth, no heuristic.
- Assuming BFS is optimal when step costs differ (you need UCS).

MY NOTES

2. Informed & Heuristic Search

Add an estimate of the remaining cost. Whether A stays optimal depends entirely on that estimate never overshooting.*

KEY FORMULAS

A* evaluation

$$f(n) = g(n) + h(n)$$

Admissible

$$h(n) \le h^*(n) \quad \text{(never overestimates)}$$

Consistent (monotone)

$$h(n) \le c(n, n') + h(n')$$

WHAT YOU MUST KNOW

- A* with an **admissible** h is optimal for tree search; with a **consistent** h it is optimal for graph search. Consistent $\Rightarrow$ admissible.
- $h \equiv 0$ turns A* into uniform-cost search. **Greedy** uses h alone and is neither optimal nor complete.
- $\max(h_1, h_2)$ of two admissible heuristics is admissible and **dominates** both — a dominating heuristic expands fewer nodes.
- $|h_1 - h_2|$ is always admissible (it is at most $\max(h_1, h_2) \le h^*$); $h_1 + h_2$ and $h_1 h_2$ are *not*.

HOW TO APPROACH A QUESTION

1. To test admissibility of a combination, ask: can it ever exceed the true remaining cost?
2. Sums and products of admissible heuristics can overshoot — max and absolute difference cannot.
3. For node-expansion order, compute $f = g + h$ at each frontier node and expand the smallest.

WORKED EXAMPLE

Q. h_1, h_2 are admissible. Which of $h_1 + h_2$, $h_1 h_2$, h_1 / h_2, $|h_1 - h_2|$ is *always* admissible?
Sums and products can exceed h^*; a ratio is unbounded. But $|h_1 - h_2| \le \max(h_1, h_2) \le h^*$:

$$\mathbf{|h_1 - h_2|}$$

WATCH OUT

- Assuming adding two admissible heuristics keeps admissibility — it usually breaks it.
- Confusing admissible (never overestimates) with consistent (the triangle inequality).

MY NOTES

3. Adversarial Search & Games

Two players with opposite goals. Propagate values up the tree; alpha-beta only skips work, never changes the answer.

KEY FORMULAS

Minimax value

$$\text{MAX node} = \max(\text{children}), \qquad \text{MIN node} = \min(\text{children})$$

Complexity

$$\text{minimax } O(b^d); \qquad \text{alpha–beta best case } O(b^{d/2})$$

WHAT YOU MUST KNOW

- Minimax assumes the opponent plays **optimally** — never that they blunder.
- Alpha–beta returns **exactly** the same root value as full minimax; it only prunes branches that cannot matter.
- α is the best value MAX can guarantee so far; β the best for MIN. Prune when $\alpha \geq \beta$.
- Pruning power depends heavily on **move ordering**; with ideal ordering you can search twice as deep.

HOW TO APPROACH A QUESTION

1. Evaluate the tree **bottom-up**, writing each node's value as you go.
2. Alternate max / min by level, starting from whichever player moves at the root.
3. If asked for the best *strategy* (not value), report which child achieved the root's value.

WORKED EXAMPLE

Q. MAX at the root has three MIN children with leaf sets $\{3,5\}$, $\{2,9\}$, $\{0,7,4\}$. Root value and best move?

$$\text{MIN nodes} = 3,\ 2,\ 0 \ \Rightarrow\ \text{root} = \max(3,2,0) = \mathbf{3}$$

MAX should play the **first** branch.

WATCH OUT

- Taking the max at a MIN level (always check whose turn each level is).
- Believing alpha–beta can change the computed value — it cannot.

MY NOTES

4. Propositional Logic

Entailment, validity and satisfiability are three phrasings of one idea. Learn how they convert into each other.

KEY FORMULAS

The equivalence chain

$$X \models Y \iff (X \Rightarrow Y) \text{ is valid} \iff (X \wedge \neg Y) \text{ is unsatisfiable}$$

Implication and contrapositive

$$P \Rightarrow Q \;\equiv\; \neg P \vee Q \;\equiv\; \neg Q \Rightarrow \neg P$$

De Morgan

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q, \qquad \neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

WHAT YOU MUST KNOW

- **Valid** = true in every model; **satisfiable** = true in at least one; **unsatisfiable** = true in none.
- Entailment runs *one way*: $X \models Y$ does **not** give $Y \models X$.
- CNF is a conjunction of clauses; resolution refutation proves $KB \models \alpha$ by deriving the empty clause from $KB \wedge \neg\alpha$.
- n propositional symbols give a truth table with 2^n rows.

HOW TO APPROACH A QUESTION

1. Rewrite the claim using the equivalence chain — most MSQ options are just restatements of it.
2. An option that reverses the direction ("if Y then X") is the standard wrong answer.
3. For a small number of symbols, a truth table settles everything.

WORKED EXAMPLE

Q. If X entails Y, which hold? (a) $X \Rightarrow Y$ (b) $X \wedge \neg Y$ is false (c) if X then Y (d) if Y then X
(a), (b), (c) are the three standard restatements of entailment. (d) reverses it.

(a), (b), (c)

WATCH OUT

- Accepting the converse "if Y then X" as equivalent.
- Confusing satisfiable with valid.

MY NOTES

5. First-Order (Predicate) Logic

Quantifiers add power and traps. Only one direction between forall and exists is valid.

KEY FORMULAS

Negation rules

$$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x), \qquad \neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$$

The one valid direction

$$\forall x\, P(x) \;\Rightarrow\; \exists x\, P(x) \quad \text{(non-empty domain)}$$

Standard translations

$$\text{all S are P}: \ \forall x(S \Rightarrow P); \qquad \text{some S are P}: \ \exists x(S \wedge P)$$

WHAT YOU MUST KNOW

- $\exists x\, P(x) \Rightarrow \forall x\, P(x)$ is **invalid** — one example does not make it universal.
- $\forall x\, P(x) \Rightarrow \exists x\, \neg P(x)$ is **invalid** — it contradicts itself.
- **Quantifier order matters**: $\exists y \forall x\, P(x, y)$ is stronger than $\forall x \exists y\, P(x, y)$ and implies it.
- "All" pairs with $\Rightarrow$; "some" pairs with $\wedge$. Swapping them is the classic translation error.

HOW TO APPROACH A QUESTION

1. To disprove validity, build a **small counter-model** — a two-element domain is nearly always enough.
2. To check a negation, push the $\neg$ inward and flip each quantifier.
3. Read $\forall/\exists$ order left to right: the later variable may depend on the earlier one.

WORKED EXAMPLE

Q. Which are NOT valid? (a) $\forall xP \Rightarrow \exists xP$ (b) $\exists xP \Rightarrow \forall xP$ (c) $\exists xP \Leftrightarrow \forall xP$ (d) $\forall xP \Rightarrow \exists x\neg P$

(a) is valid. Counter-model for (b),(c): domain $\{1, 2\}$ with $P(1)$ true, $P(2)$ false. (d) asserts a contradiction whenever $\forall xP$ holds.

(b), (c), (d)

WATCH OUT

- Writing "all S are P" as $\forall x(S \wedge P)$ — that claims everything is an S.
- Assuming $\forall\exists$ and $\exists\forall$ are interchangeable.

MY NOTES

6. Reasoning under Uncertainty

A Bayesian network is a factorised joint distribution. Read independence off the graph shape, then multiply along the arrows.

KEY FORMULAS

Factorisation

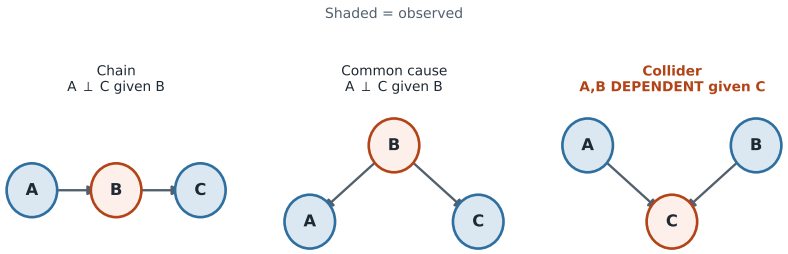
$$P(X_1, \ldots, X_n) = \prod_i P\big(X_i \mid \text{parents}(X_i)\big)$$

Bayes' rule

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

WHAT YOU MUST KNOW

- **Chain** $A \to B \to C$ and **common cause** $A \leftarrow B \to C$: A and C are independent *given* B.
- **Collider** $A \to C \leftarrow B$: A, B are independent *until* C is observed, then they become dependent ("explaining away").
- Exact inference = **variable elimination** (sum out the non-query, non-evidence variables one at a time).
- Approximate inference: rejection sampling, likelihood weighting, Gibbs/MCMC. With n variables and $\leq k$ parents you store $n2^k$ numbers instead of $2^n - 1$.



HOW TO APPROACH A QUESTION

1. For a joint probability, walk the arrows and multiply one conditional-probability entry per variable.
2. Read each factor straight off the CPT for that node's parent configuration.
3. For independence questions, classify the path shape: chain, common cause, or collider.

WORKED EXAMPLE

Q. $P(U{=}1) = 0.5$, $P(V{=}1 \mid U{=}1) = 0.5$, $P(W{=}1 \mid U{=}1) = 1$, $P(Z{=}1 \mid V{=}1, W{=}1) = 0.5$. Find $P(U{=}1, V{=}1, W{=}1, Z{=}1)$.

$$0.5 \times 0.5 \times 1 \times 0.5 = \mathbf{0.125}$$

One factor per node, each conditioned on that node's parents — nothing else is needed.

WATCH OUT

- Treating a collider like a chain and declaring independence after observing the effect.
- Multiplying marginals instead of the conditionals given the parents.

MY NOTES